MAPD Partnership Landscape — USA-Centric Proposal

Metatron Dynamics — Internal Strategy Document March 2026

Metatron's Position

Metatron Dynamics is not a hardware manufacturer. Metatron is the **system integrator and domain governance layer** for modular anthropogenic processing.

Every company listed below manufactures one type of module. No single entity consistently integrates them. Nobody declares the local feedstock domain, configures the appropriate module combination, and enforces operational boundaries across the integrated system. That is the gap Metatron fills — and it maps directly to the Origin role applied at the business architecture level.

Metatron's value proposition to each partner: **we bring you deployments you wouldn't otherwise get**, because we solve the configuration and governance problem that stands between their module and a customer's specific site.

The following companies represent potential strategic alignment targets based on publicly available information and observed operational fit. No commercial relationships or endorsements are implied.

Tier 1 — Priority US Partners (Highest Strategic Alignment)

Pyrolysis: Resynergi

Location: California (relocating from Rohnert Park for scale-up) **Why they matter:** Resynergi has developed systems that closely align with the MAPD concept — a patented Continuous Microwave Assisted Pyrolysis (CMAP) system that fits on an 18-wheeler truck. Their tagline is "No factory or facility required = fast, low risk deployment." They bring the system to the plastic waste at Material Recovery Facilities. Their manufacturing partner is Lummus, a global leader in process technology since 1907.

Strategic fit: Resynergi has the mobile pyrolysis unit. Metatron has the domain declaration and configuration framework. Together: Metatron characterizes the local feedstock (Operator A), Resynergi deploys the matched processing unit, and the ABRCE framework governs runtime operations. Resynergi is currently seeking deployment partners — they have a waitlist for CMAP system purchases and operator partnerships.

Action: Direct outreach. Their decentralized model is the MAPD pyrolysis module.

Pyrolysis: Nexus Circular

Location: Atlanta, Georgia **Why they matter:** Nexus Circular is ISCC PLUS certified and operates at commercial scale, converting landfill-bound plastics (HDPE, LDPE, PP, PS — the same four polyolefin types the MAPD sorts for) into liquid hydrocarbons for virgin-quality new plastics. They accept feedstock without costly pre-treatment, have diverted millions of pounds from landfill, and their output exceeds stringent customer quality requirements without post-treatment.

Strategic fit: Nexus has proprietary pyrolysis technology optimized for the exact feedstock streams the MAPD sorting module produces. They need characterized, sorted feedstock — which is precisely what the MAPD Module 1 delivers. Partnership model: Metatron deploys sorting + characterization depots that feed Nexus processing facilities, or co-locates Nexus processing with MAPD sorting at larger sites.

Action: Explore feedstock supply partnership. Metatron's sorting depots become Nexus's upstream supply chain.

Sorting: BHS / National Recovery Technologies (NRT)

Location: Eugene, Oregon (BHS) / Nashville, Tennessee (NRT) **Why they matter:** BHS is the leading US-based recycling systems integrator. NRT holds 35+ patents in optical sorting and their SpydIR-HS system offers hyperspectral imaging at nearly 10x standard detection resolution. In January 2026, BHS opened a new customer testing and evaluation center in Nashville for real-time material stream testing. They design complete MRF systems from engineering through installation.

Strategic fit: NRT/BHS provides the Operator A hardware — the sorting/characterization technology that declares the feedstock domain. Their systems are modular and can be configured for different waste streams. Partnership model: Metatron specifies the sorting configuration based on site characterization (domain declaration), BHS/NRT supplies and installs the sorting module, and the MAPD framework governs the integrated system.

Action: Technical partnership for Module 1 specification and supply. BHS's Nashville test center could validate MAPD sorting configurations.

Metals: Inductotherm Corporation

Location: Rancocas, New Jersey **Why they matter:** Inductotherm is the world's largest manufacturer of induction melting systems — over 36,500 systems installed in 133 countries since 1953. US-headquartered and manufactured. They cover the full range from small crucible furnaces to industrial scale, processing aluminum, copper, steel, zinc, precious metals, and specialty alloys. They offer comprehensive training programs and aftermarket support.

Strategic fit: Inductotherm provides the Module 3 (metals reprocessing) hardware at any scale the MAPD requires. Their US manufacturing base and global support network make them the natural partner for domestic deployments. Partnership model: Metatron specifies the metals processing configuration based on local scrap characterization, Inductotherm supplies the matched induction furnace system.

Action: Technical partnership for Module 3 specification. Their New Jersey headquarters is accessible for initial conversations.

Tier 2 — Strong US Candidates

Pyrolysis: Agilyx

Location: Tigard, Oregon **Profile:** Environmental technology company founded in 2004. Converts difficult-to-recycle waste plastics into crude oil, fuels, and virgin-like plastics. Partnered with ExxonMobil. Scalable commercial plants with closed-loop recycling. Expanding to Europe and Asia.

Fit: Larger scale than entry-level MAPD, but potential partnership for mid-scale deployments and downstream output off-take. Their ExxonMobil relationship could open refinery pathways for MAPD pyrolysis oil.

Pyrolysis: Plastic2Oil (P2O)

Location: Niagara Falls, New York **Profile:** Technology to convert waste plastic into liquid fuels and dirty fuel into clean diesel. Local installation model with demonstrated economic and environmental benefits.

Fit: Smaller US company with proven local-deployment model. Could partner on entry-level MAPD configurations in the Northeast.

Pyrolysis: Klean Industries

Location: Vancouver, BC (North American operations; partners with US-based KBR) **Profile:** Over 1,000 projects in 50+ countries. 100+ patents. Recently formed strategic

alliance with KBR (Houston) for global deployment of tire pyrolysis technology. Offers design-build projects from feasibility through operation.

Fit: Klean's KBR alliance gives access to US industrial engineering infrastructure. Their experience base is the deepest in the industry. Partnership model: Metatron provides domain governance, Klean/KBR provides engineering and construction for larger MAPD deployments.

Pyrolysis: Sierra Energy

Location: Davis, California **Profile:** Waste-to-energy using gasification technology (FastOx). Converts municipal solid waste, plastics, and other materials into clean energy. Focus on maximum efficiency and minimal environmental impact.

Fit: Gasification is a parallel technology to pyrolysis for certain waste streams. Potential Module 2 alternative for MSW-heavy feedstock profiles.

Sorting: MSS Optical Sorters (CP Group)

Location: National (CP Group is US-based) **Profile:** Over 50 years in automated optical sorting for recycling. NIR and AI-based systems. Serves single-stream, dual-stream, C&D, MSW, and RDF applications. CP Group designs and manufactures complete recycling systems.

Fit: Alternative or complement to BHS/NRT for Module 1. CP Group's full-system design capability aligns with MAPD integration needs.

Tier 3 — Downstream Off-Take Partners

These are not module suppliers — they are the buyers of MAPD output products. Establishing off-take relationships de-risks MAPD deployment economics.

Phillips 66

Location: Houston, Texas (Sweeny Refinery in Old Ocean, Texas) **Relevance:** Phillips 66's Sweeny Refinery achieved ISCC PLUS certification in 2024 specifically to process waste plastic pyrolysis oil into circular feedstocks (ethane, propane, propylene). They source pyrolysis oil from third-party suppliers. MAPD pyrolysis oil is exactly what they're certified to receive.

Action: Off-take agreement for MAPD pyrolysis oil production. Sweeny Refinery is a ready buyer.

Chevron Phillips Chemical

Location: The Woodlands, Texas **Relevance:** Joint venture (50/50 Phillips 66 and Chevron). Advancing circular economy for plastics. Receives circular feedstocks from Phillips 66's pyrolysis oil processing.

Fit: Second-tier off-take through the Phillips 66 pathway.

ExxonMobil (Baytown Advanced Recycling)

Location: Baytown, Texas **Relevance:** Operates advanced recycling facility processing approximately 36,000 metric tons of plastic waste per year. Actively acquiring pyrolysis feedstock capacity.

Action: Off-take or feedstock supply partnership for Gulf Coast MAPD deployments.

Tier 4 — Waste Management / Feedstock Supply Partners

These companies control the feedstock. They are both potential customers (deploying MAPDs at their sites) and supply chain partners.

Waste Management, Inc.

Location: Houston, Texas **Relevance:** Largest US waste management company. Operates hundreds of landfills and transfer stations — each of which sits on an anthropogenic reservoir.

Opportunity: Deploy MAPD units at Waste Management transfer stations and landfill sites. Convert disposal cost centers into processing revenue centers.

Republic Services

Location: Phoenix, Arizona **Relevance:** Second-largest US waste management company. Operates polymer centers and recycling facilities.

Opportunity: Same model as Waste Management — MAPD deployed at Republic facilities.

Casella Waste Systems

Location: Rutland, Vermont **Relevance:** Regional waste services company serving the Northeast. Operates landfills, transfer stations, and recycling facilities. Smaller scale, closer to entry-level MAPD deployment profile.

Opportunity: Potentially better initial partner than the majors — more agile, regional scale matches MAPD entry-level deployment.

Partnership Architecture

In every partnership, Metatron's role is consistent:

Function	Metatron's Role	Partner's Role
Site Characterization (Operator A)	Declares the feedstock domain — composition, volume, contamination profile, seasonal variation	Provides site access and operational data
Configuration Selection (Operators B + R)	Specifies module combination matched to declared domain; defines relational structure between modules	Supplies specified hardware modules
Pre-Deployment Derivation (Operator C)	Derives operational properties — break-even, energy balance, output mix, scaling limits — before deployment	Validates technical specifications
Runtime Governance (Operator E)	Provides enforcement framework for domain closure during operation	Operates and maintains equipment

Metatron does not compete with any of these companies. Metatron makes their products deployable with structural guarantees they cannot currently offer alone.

Recommended First Moves

1. **Resynergi** — Direct outreach. Their mobile CMAP system is the closest existing hardware to the MAPD pyrolysis module. They're actively seeking deployment partners. California proximity to Metatron's base.
2. **BHS/NRT** — Technical conversation about sorting module specification for MAPD. Their Nashville test center offers a validation pathway.
3. **Phillips 66 Sweeny** — Off-take inquiry. Establishing a downstream buyer for pyrolysis oil de-risks every upstream deployment decision.
4. **Casella Waste Systems** — Explore as a Northeast pilot deployment partner. Regional scale, less corporate bureaucracy than the majors, and they operate the kind of transfer stations where entry-level MAPDs fit.

5. **Inductotherm** — Technical specification for Module 3. Their New Jersey HQ is a single conversation away from module configuration for MAPD-scale aluminum and copper reprocessing.

What We're Not Doing

We are not building hardware. We are not competing with pyrolysis manufacturers, sorting equipment companies, or furnace suppliers. We are not a waste management company.

We are the entity that declares the domain, configures the system, and enforces the boundaries — so that every partner's technology operates within its valid operating envelope, every deployment is matched to its actual feedstock, and every network node is independently viable and structurally governed.

That is what an Origin does — the role responsible for declaring the domain and ensuring system validity. Applied to an industry.